

Development Guide

This document explains how to build and run the **Hypertension Risk Assessment** web application using Docker.

Prerequisites

Tool	Minimum version
Docker	24+
Git	any

Repository Layout

```
├─ container/
│   └─ Dockerfile           # Docker build definition
├─ src/
│   ├── requirements.txt
│   └─ thesis_webapp/
│       ├── app.py          # Streamlit entry point
│       ├── backend.py      # Model loading, inference,
│       │                   preprocessing
│       ├── app_constants.py # Feature metadata (labels, units,
│       │                   overrides)
│       ├── counterfactuals.py # CARLA-style counterfactual engine
│       ├── explainability.py # SHAP / LIME computation
│       └─ styles.py         # Global CSS applied to the Streamlit
app
```

Training notebooks and saved model bundles live under `src/training/`.

Build

Run the following from the **repository root** (the `Dockerfile` is inside `container/` and uses `../src/` as its copy source):

```
docker build -f container/Dockerfile -t hypertension-app .
```

The image uses `python:3.11-slim`, installs the system libraries required by LightGBM/XGBoost/OpenMP, and sets the Streamlit entry point to `thesis_webapp/app.py`.

Run

```
docker run -p 8501:8501 hypertension-app
```

Open <http://localhost:8501> in your browser.

Model Bundles

The application expects a `best_model_bundle.joblib` saved by the training notebooks. Each bundle is a `joblib`-serialised dict containing:

Key	Contents
<code>model</code>	Fitted scikit-learn <code>Pipeline</code> (preprocessor + classifier)
<code>calibrator</code>	Fitted <code>VennAbers</code> calibrator
<code>preprocessor</code>	Fitted <code>ColumnTransformer</code> (also embedded in the pipeline)

The bundle search order (highest `combined` metric wins) is:

1. `src/training/exp3_knn_rf/models/best_model_bundle.joblib`
2. `src/training/exp3_xgb_ada/models/best_model_bundle.joblib`
3. `src/training/exp3_logreg_cat/models/best_model_bundle.joblib`
4. `src/training/exp3_naive_bayes/models/best_model_bundle.joblib`

The production-pinned fallback is the **EXP B XGBoost** bundle (`exp3_xgb_ada`).

Re-running Training Experiments

Open the notebooks in `src/training/` sequentially:

Notebook	Purpose
EXP 0 Create Merge Dataset.ipynb	Merge Clinical, Dietary, Anthropometric datasets
EXP A KNN_RF.ipynb	K-NN and Random Forest experiments
EXP B XGB_AdaBoost.ipynb	XGBoost and AdaBoost experiments
EXP C LogReg_CatBoost_LightGBM.ipynb	Logistic Regression, CatBoost, LightGBM
EXP D_Naive Bayes.ipynb	Gaussian and Bernoulli Naive Bayes
EXP E EDA.ipynb	Exploratory data analysis
EXP F COMPILE_RESULTS.ipynb	Compile and compare all results
EXP G sampling exp.ipynb	Sampling strategy experiments
EXP H threshold exp.ipynb	Decision threshold experiments

Each experiment notebook saves its model bundle and calibration CSV to the corresponding `exp3_*/` directory.

Troubleshooting

Symptom	Fix
COPY failed: file not found during build	Ensure you run <code>docker build</code> from the repository root , not from inside <code>container/</code> .
Streamlit shows "No model bundle found"	Ensure a <code>best_model_bundle.joblib</code> exists in one of the <code>exp3_*/models/</code> paths under <code>src/training/</code> .

Port 8501 already in use

Run `docker run -p 8502:8501 hypertension-app` and open <http://localhost:8502>.